Counting and probability exercises

All questions are original. Sampling and independence assumptions are specified. An unknown distribution is not presumed uniform.

C-E01 Two offices versus an unordered group

Choose distinct people as president and deputy from seven people, but a designated person A cannot be deputy. Count the ordered assignments. Why can the assignment count not always be divided by 2 to count unordered groups under the same restriction? Count groups that have at least one valid assignment.

C-E02 String positions and repeated keys

For ababa, count the positional length-three substrings and the distinct contents. Count positional subsequences, including the empty one. Is this the distinct-content count? Count key strings with two copies of a, two of b, and one of c by starting from identity permutations and justifying the divisor.

C-E03 Binomial coefficients and entropy

Prove Pascal's identity and

$$\left(\frac{n}{k}\right)^k \le \binom{n}{k} \le \left(\frac{en}{k}\right)^k, \qquad 1 \le k \le n.$$

Derive the entropy bound by bounding one term of a probability expansion by 1. At $n = 8, k = 2$, compare the true coefficient with the three bounds.

C-E04 Conditional sampling

Draw twice without replacement from four green and three black balls. Find the probabilities of both draws being green, the second being green, and the second being green conditional on the first being black. Test independence. Prove generally that disjoint events with positive probabilities are dependent.

C-E05 Pairwise is not mutual independence

Fair independent bits U, V define $W = U \operatorname{XOR} V$. List the outcomes and all pairwise joint distributions. Find the probability that all three equal 1, the product of those marginals, and $\operatorname{Var}(U + V + W)$. Which calculations need pairwise independence, and which need mutual independence?

Counting and probability exercises (continued)

C-E06 Bayes and observation rules

Requests have fault probability 0.01. A monitor has sensitivity 0.8 and false-positive rate 0.05. Find the posterior fault probability after one alarm, and after two alarms when the tests are independent conditional on fault status. Why do the individual test marginals alone not determine the second answer?

C-E07 Joint masses, mean, and variance

Let $A, B \in \{1, 2, 3\}$ be independent fair three-sided dice, and let $M = \max(A, B)$. Give the mass function, mean, second moment, variance, and $P(M \geq 2)$. For $X = A, Y = 4 - A$, find the covariance and sum variance. Why can the individual variances not simply be added?

C-E08 Tail sums and nonlinear expectation

A nonnegative integer variable X has masses $P(X = 0) = 1/2$, $P(X = 1) = 1/4$, and $P(X = 3) = 1/4$. Compute its mean using both masses and tails. Find its variance and check Jensen's inequality for the square function. Does the tail formula still work if X can equal -1? Give a counterexample.

C-E09 Two geometric counting conventions

Independent trials have success probability $2/5$. Let G include the first successful trial, and let F count the earlier failures. Give their masses, means, and variances. Find $P(G > 4)$ and $P(G > 7 \mid G > 3)$. Failures cost 2 ms and success costs 5 ms; find the expected total time. Handle the boundaries $p = 0$ and $p = 1$.

C-E10 Binomial table and mode

For $n = 6$ independent trials with $p = 1/3$, list all mass numerators over the common denominator 3^6. Check normalization, mean, variance, and $P(X \geq 4)$. Find the mode from adjacent ratios. If all trials instead share one Bernoulli variable, what happens to the mean, distribution, and variance?

C-E11 Union and geometric tail bounds

For $X \sim \mathrm{Bin}(8, 1/4)$, find the exact value of $P(X \geq 5)$, the union bound, and the right-ratio bound using $k = 4$. Derive the general left-ratio bound, including the conditions for $\rho < 1$ and the distinction between strict and nonstrict thresholds.

Counting and probability exercises (continued)

C-E12 Exponential bounds and quantifiers

Let X be a sum of mutually independent Bernoulli indicators, with $\mu = \sum_i p_i > 0$. Derive

$$P(X \geq t) \leq e^{-\mu} \left(\frac{e\mu}{t}\right)^t, \qquad t > \mu,$$

from Markov. Derive the centered CLRS bound for $r > \mu$ and distinguish its threshold. At $n = 1000, p = 0.01$, compare Markov, Chebyshev, and both exponential bounds for $X \geq 50$, including their assumptions. Handle $\mu = 0$.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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